

$$1) S \rightarrow OAO \mid |B| \mid BB$$

$$a) A \rightarrow C$$

$$B \rightarrow S \mid A$$

$$C \rightarrow S \mid \epsilon$$

$$C \rightarrow \epsilon$$

$$A \rightarrow C$$

$$B \rightarrow A$$

$$S \rightarrow BB$$

$$S \rightarrow OAO \mid OO \mid |B| \mid || \mid BB \mid B$$

$$A \rightarrow C$$

$$B \rightarrow S \mid A$$

$$C \rightarrow S$$

$$b) (S, S), (A, A), (B, B), (C, C)$$

$$(S, S) \quad S \Rightarrow B \quad (S, B) \vee$$

$$(S, B) \quad B \Rightarrow A \quad (S, A) \vee$$

$$(S, A) \quad A \Rightarrow C \quad (S, C)$$

$$(A, A) \quad A \Rightarrow C \quad (A, C)$$

$$(A, C) \quad C \Rightarrow S \quad (A, S)$$

$$(A, S) \quad S \Rightarrow B \quad (A, B)$$

$$(B, B) \quad B \Rightarrow B, B \Rightarrow A \quad (B, S)(B, A)(B, C) \vee$$

$$(C, C) \quad C \Rightarrow S \quad (C, S)(C, B)(C, A) \vee$$

$$c) T = \{0, 1\}$$

$$W_1 = \{S, A, B, C\}$$

$$W_2 = \{S, A, B, C\} = W_1$$

$$Y_1 = \{S\}$$

$$Y_2 = \{S, A, B, 0, 1\} \quad (= \text{inacanzabile})$$

$$Y_3 = \{S, A, B, 0, 1\} = Y_2$$

$$P: S \rightarrow OAO \mid OO \mid |B| \mid || \mid BB$$

$$A \rightarrow OAO \mid OO \mid |B| \mid || \mid BB$$

$$B \rightarrow OAO \mid OO \mid |B| \mid || \mid BB$$

d) $Z \rightarrow \emptyset$
 $O \rightarrow 1$
 $S \rightarrow ZAZ | ZZ | OBO | OO | BB$
 $C \rightarrow AZ$
 $M \rightarrow BO$

R11 $S \rightarrow ZC | Z\bar{Z} | OM | OO | BB$
 $A \rightarrow ZC | Z\bar{Z} | OM | OO | BB$
 $B \rightarrow ZC | Z\bar{Z} | OM | OO | BB$
 $C \rightarrow AZ$
 $M \rightarrow BO$
 $Z \rightarrow \emptyset$
 $O \rightarrow 1$

7) $S \rightarrow aAa | bBb | \epsilon$ $S \rightarrow \epsilon$ $S \rightarrow aAa | aa | bBb | bb$
a) $A \rightarrow C | a$ $C \rightarrow \epsilon$ $A \rightarrow C | a$
 $B \rightarrow C | b$ $A \rightarrow C$ $B \rightarrow C | b$
 $C \rightarrow CDE | \epsilon$ $B \rightarrow C$ $C \rightarrow CDE | DE | CE | E$
 $D \rightarrow A | B | ab$ $D \rightarrow A$ $D \rightarrow A | B | ab$

b) $(S, S) (A, A) (B, B) (C, C) (D, D) (E, E)$
 $(A, A) A \rightarrow C (A, C) \cup$
 $(A, C) C \rightarrow E (A, E) \cup$
 $(B, B) B \rightarrow C (B, C) (B, E) \cup$
 $(C, C) C \rightarrow E (C, E) \cup$
 $(D, D) D \rightarrow A (D, A) (D, B) (D, C) (D, E) \cup$

c) $T = \{a, b\}$

$W_1 = \{S, A, B, D\}$

$W_2 = \{S, A, B, D\} = W_1$

(y $E = \text{inútiles}$

$Y_1 = \{S\}$

$Y_2 = \{S, A, B, a, b\}$

$Y_3 = \{S, A, B, a, b\} = Y_2$

$D = \text{inalcanzable}$

P: $S \rightarrow aAa \mid aa \mid bBb \mid bb$

$A \rightarrow a$

$B \rightarrow b$

d) $A \rightarrow a$

$B \rightarrow b$

$C \rightarrow AA$

$M \rightarrow BB$

$S \rightarrow AC \mid AA \mid BM \mid BB$

③ $S \rightarrow ASA \mid aB$ $B \rightarrow \epsilon$

a) $A \rightarrow B \mid S$ $A \rightarrow B$

$B \rightarrow b \mid \epsilon$

b) $(S, S)(A, A)(B, B)$

$(A, A) \quad A \Rightarrow B \quad (A, B)^\cup$

$(A, A) \quad A \Rightarrow S \quad (A, S)^\cup$

c) $T = \{a, b\}$

$W_1 = \{S, A, B\}$

$W_2 = \{S, A, B\} = W_1$

$$Y_1 = \{S\}$$

$$Y_2 = \{S, A, B, a\}$$

$$Y_3 = \{S, A, B, a, b\}$$

$$Y_4 = \{S, A, B, a, b\} = Y_3$$

d) $X \rightarrow a \quad B \rightarrow b$

$$C \rightarrow SA$$

$$S \rightarrow AC | SA | AS | XB | a$$

$$A \rightarrow AC | SA | AS | XB | a | b$$